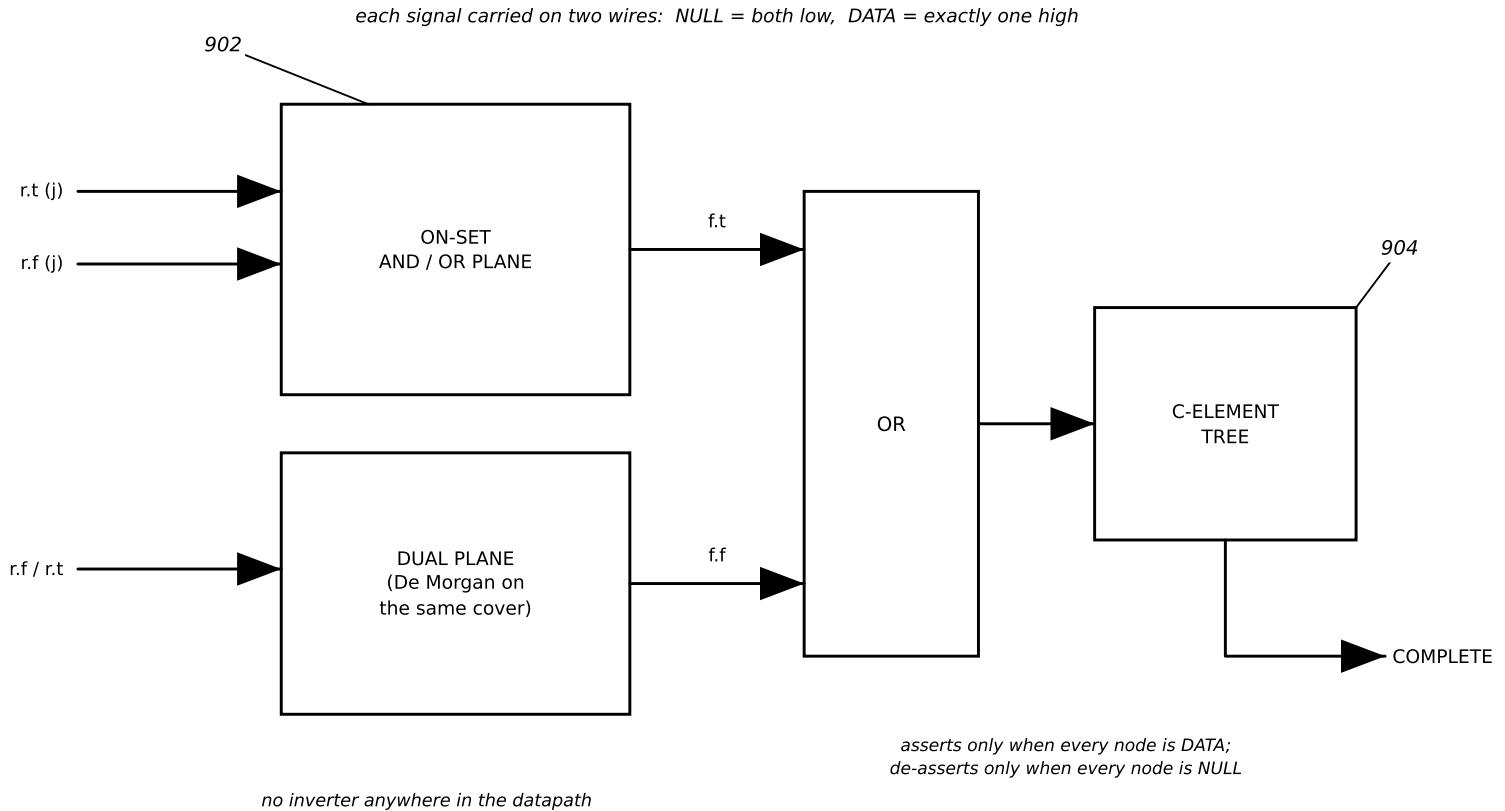


ALTERNATIVE EMBODIMENT — DELAY-INSENSITIVE NODE



WHY THIS IS HAZARD-FREE WITHOUT ANY TIMING ASSUMPTION

Out of NULL every rail rises and none falls, so every product term is a monotone function of the rails and every internal node can only go $0 \rightarrow 1$. A node that never falls cannot glitch. Correctness therefore does not depend on the sizing rule of FIG. 8.

The false rail is obtained from the true rail's cover by De Morgan, not by a second independent minimisation. Two covers minimised against the same don't-care set leave inputs covered by neither, both rails then stay low, and completion detection waits for a DATA that never arrives.

This embodiment does NOT replace the partitioning. A C-element prevents a node from acting on inputs that have not settled; it does nothing about two coupled nodes acting on the same stale state. The same dual-rail array with the partitioning removed does not converge.

It costs roughly twice the logic of FIG. 8 and one NULL phase per evaluation.

FIG. 9